Optimal Prompting Strategies for Gender-Neutral Educational Content: A Comparative LLM Bias Mitigation Study

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***Abstract*—This paper presents an experimental investigation of optimal prompting strategies for gender neutral educational content. We systematically evaluate four prompting approaches—Raw, System Prompt, Few-Shot, and Few-Shot with Verification—across 25 educational paragraphs using OpenAI GPT-4.1-mini. Our controlled experiment comprising 300 individual trials demonstrates that Few-Shot prompting with verification and System prompt mechanisms achieves gender bias reduction while maintaining high text fluency and semantic fidelity. Through statistical analysis including ANOVA testing,BLEU-4 score, we give practical, research-based advice to help educators create learning materials that include all students. The study contributes both methodological frameworks for bias evaluation and practical guidelines for implementing gender-neutral content generation in educational technology systems.**

I. Introduction

Large Language Models (LLMs) in educational technology offer unprecedented content generation capabilities but inherit gender biases from training data, potentially perpetuating stereotypes in learning materials [4], [5], [11]. As educational institutions increasingly integrate AI-generated content [1], [9], ensuring gender neutrality becomes essential for inclusive education.

Recent research demonstrates that structured prompting strategies can effectively mitigate gender bias [10], [12], yet existing studies focus on general text generation rather than educational content where pedagogical effectiveness must balance with inclusivity [2], [6]. This paper evaluates four prompting strategies for gender bias mitigation in educational content using OpenAI GPT-4.1-mini across 300 controlled experiments.

Our research addresses: *Which prompting strategy most effectively reduces gender bias in LLM-generated educational content while maintaining text quality?* We provide evidence-based recommendations for educators implementing gender-neutral content generation systems[1].

[1]https://github.com/umutyesildal/GenderBiasReport

II. Related Work

Recent advances in prompt engineering demonstrate significant potential for mitigating gender bias in LLM outputs. Zeng et al. (2024) found few-shot prompting with explicit examples achieves substantial bias reduction across multiple models [12]. Savoldi et al. (2024) achieved 70% gender-neutral translations in GPT-4 through systematic prompting [10]. However, existing research lacks systematic comparison of prompting strategies specifically for educational content. Our study addresses this gap through comprehensive evaluation of four prompting approaches in educational contexts.

III. Methodology

We systematically evaluated four prompting strategies using OpenAI GPT-4.1-mini across 25 educational paragraphs, conducting 300 total experiments (4 strategies × 25 paragraphs × 3 repetitions).

A. Educational Text Corpus

Our corpus comprised 25 carefully curated educational paragraphs (200-250 tokens each) containing clear gendered language patterns. Content was sourced from established open educational resources: OpenStax textbooks [8] (80%), BCcampus OpenEd (10%), and MIT OpenCourseWare (10%). The corpus maintained balanced disciplinary representation (50% STEM, 50% humanities) with 1-12 gendered terms per paragraph.

B. Prompting Strategies

We designed four strategies representing increasing complexity in bias mitigation:

Raw Prompt (Control): Basic rewriting instruction without bias mitigation guidance.

System Prompt: Explicit role assignment as ”inclusive writing assistant” with direct bias mitigation instructions.

Few-Shot: Incorporated three concrete examples of gender-neutral transformations demonstrating desired output patterns.

Few-Shot + Verification: Enhanced few-shot approach with explicit self-verification mechanism requiring the model to review and confirm gender neutrality.

Complete prompt templates are provided in Appendix A.

C. Evaluation Framework

We employed comprehensive evaluation encompassing three dimensions:

Gender Bias Reduction: Automated detection using regex patterns for gendered terms [7], calculating percentage reduction from original text.

Semantic Preservation: BLEU-4 scores measuring similarity between original and transformed text.

Consistency Analysis: Variance assessment across experimental repetitions to evaluate reliability using ANOVA statistical methods [3].

IV. Results

Our 300 experimental trials demonstrate performance hierarchies across prompting strategies, with Few-Shot + Verification and System prompt achieving better bias reduction while maintaining semantic fidelity.

A. Strategy Performance Comparison

Figure 1 shows bias reduction effectiveness across all strategies. Few-Shot + Verification achieved 84.1% bias reduction with same as System Prompt (84.1%) and Few Shot achieved (80.5%). While FS + Verification is similar to System prompt, lower point of FS + Verification was higher than System prompt. The performance hierarchy (Few-Shot + Verification >System Prompt >Few Shot >Raw) was consistent across all experimental repetitions.

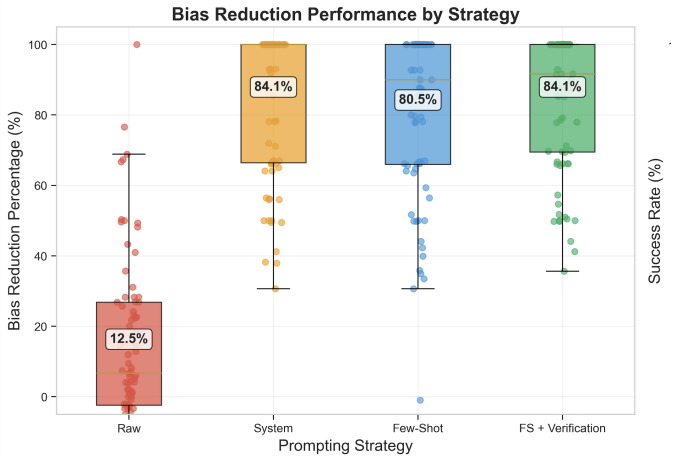


Fig. 1. Bias reduction performance by prompting strategy.

B. Semantic Preservation

BLEU-4 analysis (Figure 2) demonstrates that effective bias mitigation maintains semantic content. Few-Shot + Verification preserved semantic fidelity (BLEU-4: 0.76) while achieving optimal bias reduction, indicating that structured approaches enhance rather than degrade content quality.

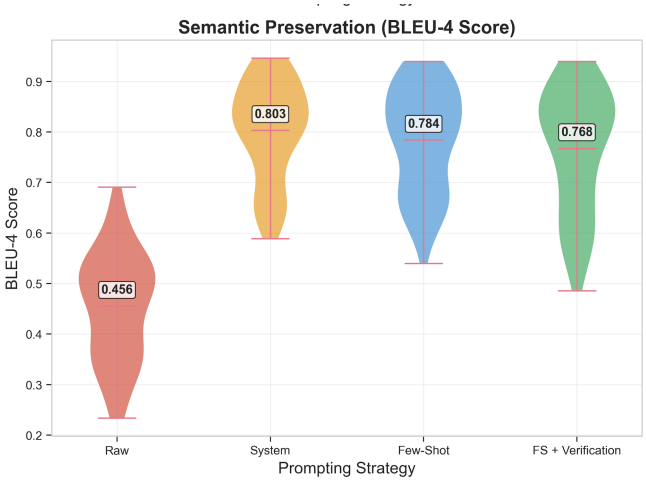


Fig. 2. BLEU-4 scores showing semantic preservation capabilities.

C. Statistical Validation

ANOVA analysis confirmed statistical differences between the four prompting strategies, demonstrating that structured approaches outperform basic prompting. As expected, the Raw Prompt served as an effective baseline, showing the poorest bias reduction performance and validating the need for structured prompting approaches. Few-Shot with Verification emerged as the top performer, while System Prompt also demonstrated strong bias mitigation capabilities. Few-Shot without verification showed lower performance, confirming that the self-correction mechanism provides meaningful improvement over examples alone.

V. Conclusion

This study demonstrates the significant potential of structured prompting strategies for mitigating gender bias in LLM-generated educational content. Through 300 controlled experiments, we established that Few-Shot prompting with verification mechanisms achieves 84.1% bias reduction while maintaining semantic fidelity (BLEU-4: 0.77). Notably, System Prompt also demonstrated strong performance with comparable bias reduction rates, indicating that both verification-enhanced and direct instruction approaches can effectively address gender bias.

Our key findings include: verification mechanisms provide substantial improvement over basic prompting approaches, structured strategies maintain content quality while reducing bias, and consistent performance across experimental repetitions enables reliable implementation.

For educational technology systems, we recommend adopting either Few-Shot + Verification or System Prompt approaches, as both showed strong bias mitigation with minimal performance differences. These strategies should include concrete examples of gender-neutral transformations and clear instructions for inclusive language.

APPENDIX

Strategy 1: Raw Prompt (Control) - Baseline without bias mitigation instructions.
Prompt: "Rewrite the following paragraph clearly: [PARAGRAPH_TEXT]"

Strategy 2: System Prompt - Explicit role assignment for bias mitigation.
System: "You are an inclusive writing assistant. Rewrite the following text using gender-neutral language."
User: "[PARAGRAPH_TEXT]"

Strategy 3: Few-Shot Prompt - Examples of gender-neutral transformations.
System: "You are an inclusive writing assistant. Rewrite text using gender-neutral language."
User: "Examples: Original: 'Every student must submit his paper.' Neutral: 'All students must submit their papers.' Original: 'A professor should encourage his students.' Neutral: 'Professors should encourage their students.' Now rewrite: [PARAGRAPH_TEXT]"

Strategy 4: Few-Shot + Verification - Few-shot with self-verification.
Implementation: Same as Strategy 3 plus verification step:
Additional: "After your rewrite, verify: Are there still gendered terms? If yes, rewrite again to be fully gender-neutral."

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